

Colour evidence that the companion of a quasar is itself a quasar at the same redshift

Method note for `qso_pcolor`

September 20, 2026

Abstract

A quasar has a spectroscopic redshift z_0 ; a companion a few arcseconds away has broadband photometry and nothing else. We want to know how strongly the companion’s colours support the hypothesis that it is a quasar at $z \simeq z_0$. This note sets out the method implemented in `qso_pcolor`: a class-conditional colour–redshift density for quasars, a local colour–magnitude–sky density for the field population, explicit surface densities that convert those into intensities, and a three-hypothesis posterior that includes quasars at the *wrong* redshift as a competing explanation. We show that the redshift-match window enters the answer as a pure multiplicative constant, derive the window-free statistic R that follows from that, and explain why R rather than a posterior probability is the quantity to rank on. All figures are built from DESI DR1 spectroscopy and Legacy Surveys DR9 imaging. Applied to those data, the statistic separates quasars from the field population by about fourteen natural logarithms but separates same-redshift from wrong-redshift quasars by only about two: these bands identify quasars well and place them in redshift only weakly, which is the central practical constraint on the method.

The method in plain words

An orientation before the formalism. Everything asserted here is developed, and qualified, in the sections that follow.

The question, and what is in scope. We have a quasar whose redshift has been measured from a spectrum, and a neighbour a few arcseconds away for which we have only photometry — a brightness in each of a handful of filters. We want to know whether that neighbour is also a quasar, at the same redshift. The pipeline scores only companions far enough from the primary that the survey photometry treats them as separate sources; closer pairs need image-level work that is not attempted here (§11).

How we learn what quasars look like. We take the quasars DESI has observed spectroscopically — about 80,000 in the patch used for the figures — and attach the brightnesses that the Legacy Surveys imaging measured for those same objects. The two catalogues share an identifier, so the photometry is attached by name rather than by position; removing objects that appear twice under different identifiers is a separate step (§10). That gives a cloud of points in colour space with a measured redshift attached to each. We fit a separate distribution in each of a set of *overlapping* redshift slices, and blend the two neighbouring distributions according to where in between a given redshift falls. The result answers: *if this object is a quasar at redshift z , how likely are these colours?*

One property of that sample must be carried forward: DESI selects quasars for spectroscopy using a classifier built on these very colours. What we learn is therefore the colour distribution of quasars *that DESI would target*, not of quasars in general.

Why measurement error is handled explicitly. Every measured colour carries an uncertainty, and those uncertainties are correlated between colours because the colours share bands. Fitting the observed cloud would give a locus broader than the true one, and would make unusual colours look less unusual than they are. So we fit the *underlying* distribution, using each training object’s full measurement covariance, and then — when scoring a candidate — add that candidate’s own covariance back in. A noisy candidate is thus compared against an appropriately blurred locus, not against an artificially sharp one.

What we use for the competing population. Not stars alone. We take every source in the same imaging that passes the same quality cuts as the candidate, and model its colours the same way: stars, faint galaxies, blends, whatever the survey actually contains. The reason is that a chance neighbour is drawn from everything that is there. Estimating that population directly from catalogue detections avoids having to reconstruct what lies below the detection limit, though it still requires that the candidate cuts and the usable sky area match. Because this population changes across the sky and with depth, we fit it separately in patches of sky and in bins of brightness, falling back to coarser patches where a cell holds too few objects. For scoring a handful of specific candidates there is a second and better option: fit the background from the candidate’s own neighbourhood (§5).

One honest complication: this sample is not currently purged of quasars, so the “quasar” and “everything else” categories overlap and part of the quasar population is counted twice. Their fraction among all sources is tiny, but that does not bound the effect where quasar colours live, which is the only place it matters. The size of it has not yet been measured (§5).

How the two are compared. Two colour distributions alone cannot answer the question. They give the relative colour evidence — useful, and reported — but not a probability, because that also depends on how many objects of each kind exist at this brightness on this part of the sky. So each distribution is multiplied by a surface density: quasars per square degree per unit brightness *per unit redshift*, and field sources per square degree per unit brightness. The quasar quantity is then summed over redshift, separately inside and outside the declared match window, which is what makes the two comparable. What come out are expected-count densities for three possibilities:

- a quasar at the primary’s redshift,
- a quasar at some *other* redshift,
- anything else in the imaging.

The middle one is not a refinement. A genuine quasar behind the primary at a quite different distance looks nothing like a star, so a two-way quasar-versus-star comparison would count it as evidence for a companion. For a pair search that is the main way of being wrong.

Why the answer is a ranking rather than a verdict. Colours locate a quasar in redshift only loosely: on our models the redshift posterior has a median standard deviation of about 0.6, and is often multimodal. A physically motivated pair window of $\pm 2000 \text{ km s}^{-1}$ spans about 0.037 in redshift at $z = 1.8$. The window is far narrower than the measurement, so only a small part of a candidate’s redshift probability can ever fall inside it, and the resulting probability is correspondingly small. Strong colour evidence can therefore coexist with a small probability of falling inside such a narrow window; a small number is not by itself a verdict against the candidate.

For that reason we report a statistic that the width of the window barely affects — exactly so in the limit of a narrow window, approximately so in practice — and convert it to a probability only at the end, for a window the user declares. Candidates are ordered by that statistic; spectroscopy decides. Note also that even a confident answer here is a statement about colours and populations,

not about whether two quasars are physically bound: that needs a clustering model this pipeline deliberately leaves out.

What the illustrative numbers do and do not show. Applied to the sky patch used for the figures, the ranking statistic puts quasars far from the general field population, and same-redshift quasars only modestly above wrong-redshift ones — medians differing by about 14 and about 2.3 natural logarithms respectively (§7.6). These are in-sample comparisons from one untuned model at one redshift, with the hardest intermediate cases absent, so they indicate the dynamic range available rather than any measured purity or completeness. Read at that level, they say that these filters identify quasars much better than they place them in redshift. Whether extra filters with a steeper colour–redshift dependence would improve matters is a reasonable expectation from §7.3, not something demonstrated here.

1 The question, and the question that can be answered

The scientific target is physical quasar pairs. The observational situation is that one member has a spectrum and the other does not, and obtaining the second spectrum is expensive. A method that ranks companions by how quasar-like their colours are *at the primary’s redshift* therefore decides where telescope time goes.

Two things must be separated at the outset, because conflating them is the standard failure of this kind of analysis.

A likelihood is not a probability. The quantity $p(\mathbf{c} \mid Q, z_0)$ — how probable these colours are *if* the object is a quasar at z_0 — says nothing on its own about whether the object is a quasar. Turning it into a probability requires knowing how many quasars and how many other objects exist at this brightness and sky position. We therefore compute and report both, and never let the first be read as the second.

Two hypotheses are not enough. A calculation that pits “quasar at z_0 ” against “star” will assign a high probability to a genuine quasar at $z = 2.6$ sitting beside a primary at $z = 1.4$, because such an object is indeed nothing like a star. For a pair search that is not a detail; it is the dominant contaminant. The competing hypotheses must include field quasars.

2 Hypotheses and intensities

Let $\mathbf{c}$ be a vector of colour/shape features, m a reference magnitude, and (ℓ, b) Galactic coordinates. We consider

H_{same}	a quasar whose redshift matches z_0 under a declared window
H_{field}	a quasar at any other redshift
H_{bkg}	anything else in the imaging catalogue at this m and (ℓ, b)

H_{bkg} is deliberately not “star”. The population a chance projection draws from is everything the imaging contains — stars, compact galaxies, unclassified blends — and that population has the decisive practical advantage of requiring no selection function, because it is exactly the set of objects being counted (§5).

We work with *intensities*: expected counts per unit sky area, per unit magnitude, per unit volume of colour space. Writing $W(z \mid z_0) \in [0, 1]$ for the redshift-match weight, $\Sigma_Q(z, m)$ for the quasar surface density and $\Sigma_B(m, \ell, b)$ for the background surface density,

$$\lambda_{\text{same}} = \int W(z \mid z_0) \Sigma_Q(z, m) p(\mathbf{c} \mid Q, z) dz, \quad (1)$$

$$\lambda_{\text{field}} = \int [1 - W(z \mid z_0)] \Sigma_Q(z, m) p(\mathbf{c} \mid Q, z) dz, \quad (2)$$

$$\lambda_{\text{bkg}} = \Sigma_B(m, \ell, b) p(\mathbf{c} \mid B, m, \ell, b). \quad (3)$$

All three carry the same units, so their ratios are meaningful, and

$$p_{\text{same}} = \frac{\lambda_{\text{same}}}{\lambda_{\text{same}} + \lambda_{\text{field}} + \lambda_{\text{bkg}}}. \quad (4)$$

Alongside this we always report the prior-independent colour Bayes factor

$$\log \text{BF} = \log p(\mathbf{c} \mid Q, z_0) - \log p(\mathbf{c} \mid B, m, \ell, b), \quad (5)$$

which depends on the two density models but on neither surface density. If the surface densities change, (4) must move and (5) must not; this is enforced by a unit test rather than by intention.

3 Photometric features and their covariance

3.1 Relative fluxes

The default feature vector divides each band’s flux by a reference band, $r_j = f_j/f_a$, giving a representation of spectral shape that is independent of overall brightness. Two reasons make fluxes rather than magnitudes the working variable.

A faint source can have a negative measured flux. That is a real measurement — the true flux is small and the noise took it below zero — and it carries information about the spectral energy distribution. Its magnitude is undefined. In the samples used here, 21% of the background sources that enter the fit have at least one negative flux ratio, almost entirely in WISE (18.7% in f_{W2}/f_r , 7.5% in f_{W1}/f_r); the figure for quasars is 1.35%. Those are not marginal objects to be discarded: faintness in WISE is one of the strongest signals that something is *not* a quasar, so dropping them, or replacing them with upper limits, would remove the most informative part of the competing population and bias the background model towards mid-infrared-bright sources. Working in flux ratios keeps them, and the observed-dimension mask is reserved for bands the survey itself declares unusable — zero inverse variance, zero exposures, a mask bit — never for a band that merely came out faint or negative.

The errors are Gaussian in flux. The imaging pipeline’s noise model is Gaussian in flux, and every density in this package — the mixture components, the error convolution, the extreme-deconvolution fit — assumes Gaussian measurement errors. A logarithm makes that assumption false exactly where it matters, turning a symmetric flux error into a skewed, heavy-tailed magnitude error at low signal-to-noise. This is the same reasoning that led XDQSO to model relative fluxes (Bovy et al., 2011b).

What the choice costs. The denominator is itself noisy, so (6) is a linearisation that degrades once the reference band is not significantly positive; objects below a configurable reference signal-to-noise are flagged rather than silently trusted. At the depth used here that is rare — 0.03% of the fitted background and 0.31% of the quasars — but it would not be at a fainter limit. The more substantial cost is distributional: flux ratios have long positive tails, with f_{W1}/f_r reaching several tens (Figure 4), and a Gaussian mixture needs more components to represent a heavy tail than it would in a compressed representation. The asinh (“luptitude”) colours implemented alongside are the natural alternative: they are finite for negative and zero flux, so they keep the first advantage, while compressing the tails. Which representation predicts held-out colours better has not been tested, and is one of the open questions.

The covariance follows by the Jacobian. With independent band fluxes of variance v_j ,

$$[\mathbf{C}_r]_{jk} = \frac{\delta_{jk} v_j}{f_a^2} + \frac{r_j r_k v_a}{f_a^2}, \quad (6)$$

whose second term is a dense, strictly positive contribution from the shared denominator. Treating feature errors as independent would therefore be wrong even when the band fluxes are independent. The same applies to ordinary colours, which share bands by construction: for $\mathbf{c} = \mathbf{A}\mathbf{m}$ the covariance is $\mathbf{A}\mathbf{C}_m\mathbf{A}^\top$, and adjacent colours are anticorrelated through the band they share.

3.2 Extinction

Fluxes are dereddened by the per-object transmission, $f \rightarrow f/T$, and the correction is propagated into the variance as $v \rightarrow v/T^2$. At low Galactic latitude the correction is large, and carrying it in the flux while leaving the variance alone would understate the uncertainty precisely where it is greatest.

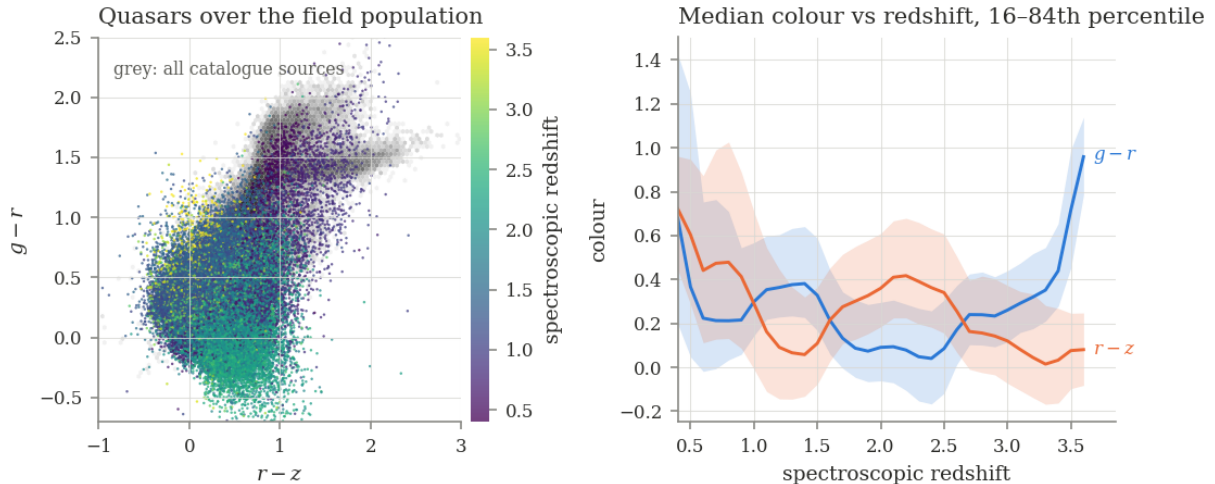


Figure 1: DESI DR1 quasars in Legacy Surveys DR9 photometry. *Left*: the quasar locus, coloured by spectroscopic redshift, over the density of all catalogue sources in the same field (grey). The quasar and field populations overlap substantially — discrimination comes from where in the plane an object sits, not from a clean separation. *Right*: median colour against redshift with the 16–84th percentile band. The track is strongly non-monotonic, which is the property that shapes the choice of model in §4.3.

4 The quasar colour–redshift model

4.1 What the data look like

Figure 1 shows the problem. Quasars trace a track through colour–colour space that moves with redshift, doubles back on itself, and crosses the locus of ordinary catalogue sources more than once. The motion is driven by strong emission lines and the Lyman break entering and leaving the passbands, so it is neither monotonic nor smooth on the scale of a passband width.

4.2 Extreme deconvolution

Each object’s colours are measured with its own heteroscedastic, correlated error. Fitting a mixture to the observed points would recover the locus *convolved* with those errors, which is systematically too wide, and the width is exactly what discriminates. We therefore fit the *intrinsic* distribution by extreme deconvolution (Bovy et al., 2011): a Gaussian mixture

$$p(\mathbf{v}) = \sum_k \alpha_k \mathcal{N}(\mathbf{v} \mid \boldsymbol{\mu}_k, \mathbf{V}_k) \quad (7)$$

whose components are convolved with each observation’s covariance $\mathbf{S}_i$ during fitting, so that the density of the *observed* value is evaluated with $\mathbf{V}_k + \mathbf{S}_i$. The EM update is given in Appendix B. Figure 2 tests this on real quasar colours by injecting noise of known size: the deconvolved width is recovered to within 7% after the observed width has more than doubled, while an ordinary mixture tracks the broadened distribution exactly. The correction matters most where the measurement error approaches the intrinsic width of the locus, which in these data is the faint end.

4.3 Conditional slices rather than a joint mixture

There are two natural ways to make the mixture redshift-dependent.

The first, following XDQSOz (Bovy et al., 2012), fits a single mixture to the joint vector $[\mathbf{c}, z]$ and conditions analytically on z . It is elegant and gives a closed-form photometric redshift distribution. It has two properties that are awkward here. Within a component the colour–redshift

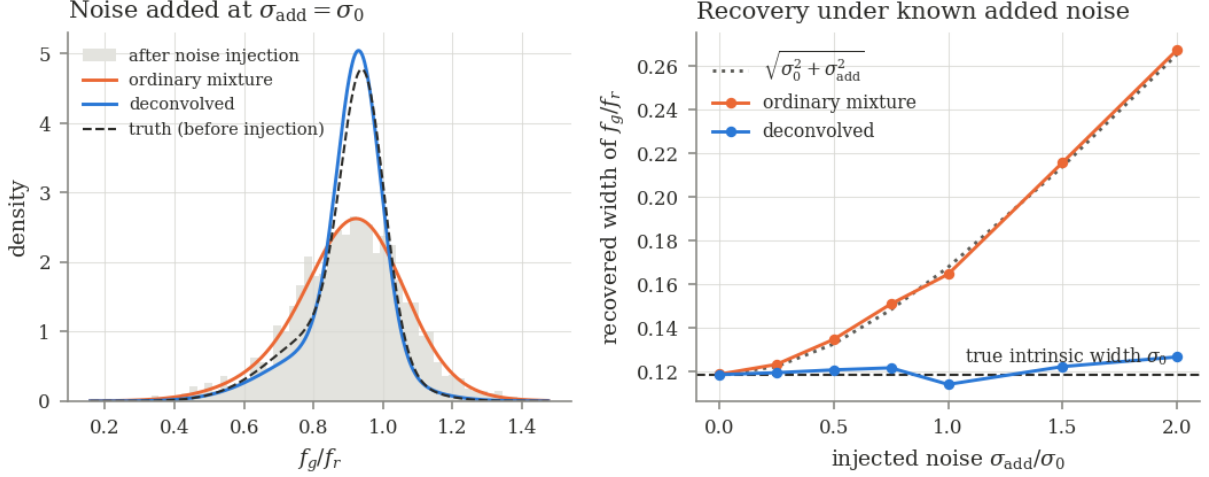


Figure 2: Deconvolution tested by controlled noise injection on real quasar colours. Bright quasars near $z = 1.8$ ($N = 1536$, $r < 20$) have negligible measurement errors, so their fitted width $\sigma_0 = 0.119$ is effectively the intrinsic one. Gaussian noise of known size is then added and the models refitted. *Left*: at $\sigma_{\text{add}} = \sigma_0$ the deconvolved mixture (blue) recovers the pre-injection truth (dashed) while the ordinary mixture (orange) is visibly broadened. *Right*: the ordinary mixture follows $\sqrt{\sigma_0^2 + \sigma_{\text{add}}^2}$ exactly, as it must, reaching 0.267 at $\sigma_{\text{add}} = 2\sigma_0$; extreme deconvolution stays flat, returning 0.127 — within 7% of the truth after the observed width has more than doubled.

relation is *linear*, so representing the strongly curved track of Figure 1 requires components narrow in z , and the required number is set by the curvature rather than by the data volume. Second, conditioning on z reweights components by their z -marginal, which is the redshift distribution of the *training sample* — that is, of the spectroscopic target selection. The conditional density is still correct, but the redshift prior enters implicitly.

The second, which we adopt as the default, fits an independent deconvolved mixture in each of a set of overlapping redshift slices and interpolates linearly in density between neighbouring slice centres:

$$p(\mathbf{c} \mid Q, z) = (1 - w)p_j(\mathbf{c}) + wp_{j+1}(\mathbf{c}), \quad w = \frac{z - z_j}{z_{j+1} - z_j}. \quad (8)$$

Because each p_j is normalised over colour space, (8) is a normalised density for any z , and the training sample’s redshift distribution *between* slices cannot leak into it. It does not follow that no redshift-selection effect survives: a slice of finite width estimates the colour distribution averaged over the training redshifts *within* that slice, so a strongly varying targeting rate inside a slice still biases it, and colour-dependent selection affects both this and the joint construction equally. Narrower slices reduce the residual at the cost of statistics.

What the slice construction does buy is that the redshift resolution becomes an explicit, tunable quantity, and that the redshift prior is supplied separately and visibly as $\Sigma_Q(z, m)$ rather than inherited.

Figure 3 shows the fitted conditional as contours in one plane at three redshifts; Figure 4 shows the same model as a function of redshift, in each of the four features, against the data it was trained on. Figure 3 shows the fitted conditional at three redshifts.

Both backends implement the same density interface, but they are not yet interchangeable in the scorer, which calls a slice-specific method to reuse precomputed per-slice densities. Using the joint model for candidate scoring needs that path generalised first. Which predicts held-out quasar colours better remains an empirical question, to be settled by measurement rather than by the argument above.

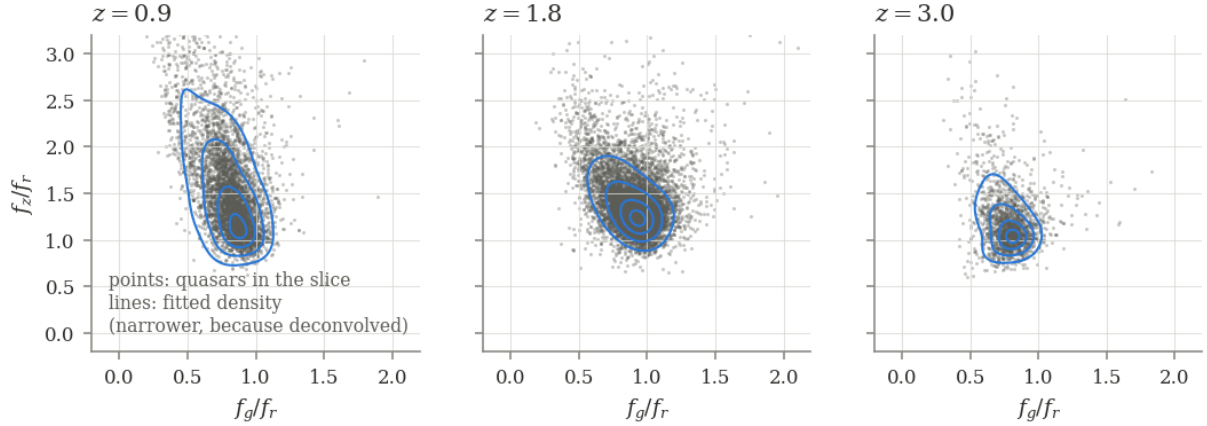


Figure 3: The fitted conditional density $p(\mathbf{c} \mid Q, z)$ at three redshifts (contours), with the quasars in each slice (points). The locus translates and changes shape with redshift; each slice is a separate deconvolved mixture, so no linear colour–redshift relation is imposed. The contours are visibly tighter than the scatter of the points because the model is the *deconvolved* density, while the points carry their measurement errors (§4.2). Note also how much the three loci overlap: this is the origin of the weak redshift discrimination quantified in §7.6.

4.4 Missing bands

A Gaussian mixture marginalises exactly over unobserved dimensions: one deletes the corresponding rows and columns of every component covariance and of the observation’s covariance, and evaluates the resulting lower-dimensional density. No imputation is involved, and the result is the correct likelihood of what was actually measured. The practical consequence is that a candidate lacking one band is still scored, with correspondingly weaker discrimination, rather than being dropped or given a survey-median colour.

5 The background model

The competing density $p(\mathbf{c} \mid B, m, \ell, b)$ is built from the imaging catalogue itself, over the same footprint and with the same quality definition applied to the candidates. No star/galaxy separation is imposed, because imposing one would require a selection function we do not have, whereas the unfiltered catalogue *is* the population a chance projection draws from.

Known quasars are removed, and it matters more than it looks. The formalism of §2 treats H_{field} and H_{bkg} as mutually exclusive, so quasars must not appear in both. The tempting argument for ignoring this is that quasars are a tiny fraction of all catalogue sources. That is true and beside the point: what matters is their fraction *in quasar-like colour space*, which is exactly where the same-redshift hypothesis is tested. Measured on a one-degree cone:

log BF (quasar at z_0 / background)	sources	known quasars
−40 to −10	17,995	0.1%
−5 to −2	6,026	0.5%
0 to +2	978	9.5%
+2 to +5	725	31.2%
+5 to +40	272	72.4%

Overall the sample is 1.0% quasars; where a genuine companion would sit it is 72%. Left in, the background model learns the quasar locus and competes against the quasar hypothesis

The fitted quasar model across redshift, in every feature it uses

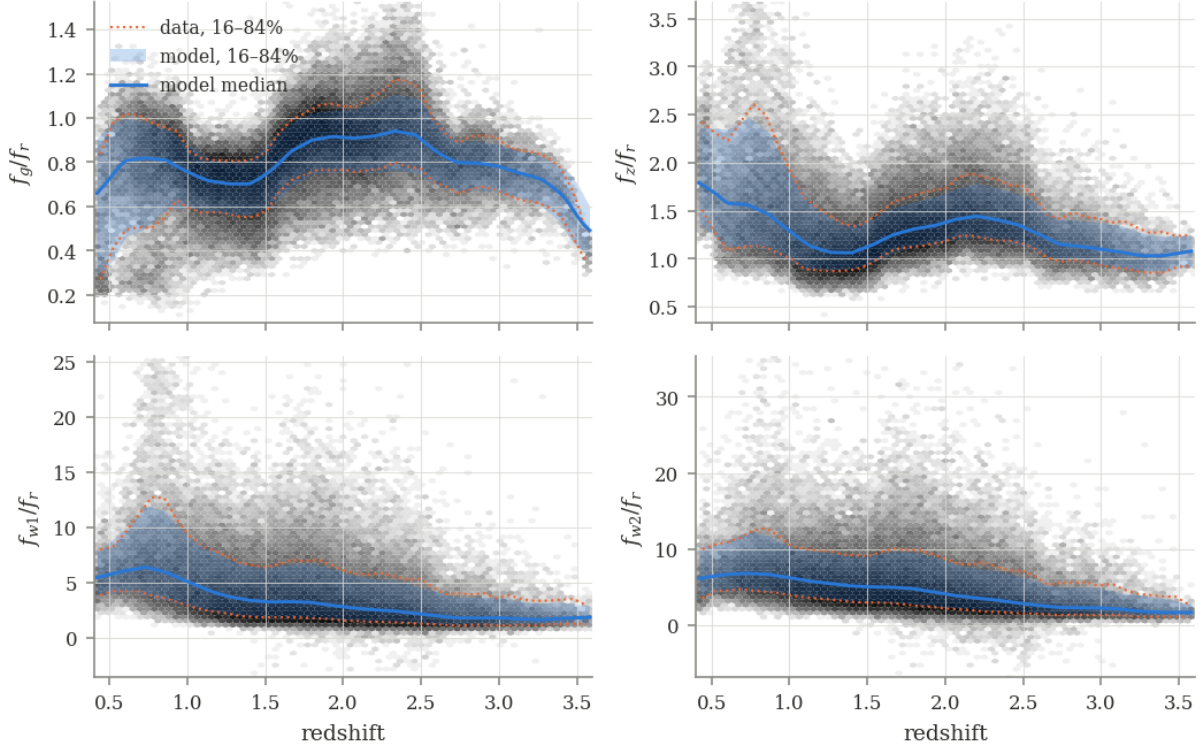


Figure 4: The fitted quasar model itself, over the whole trained redshift range and in every feature it uses. Grey is the training data; the blue band and line are the 16–84th percentile and median of the *model’s* marginal distribution at each redshift, computed from the fitted mixtures; the dotted orange lines are the same percentiles of the data. Figure 3 shows this density as contours in one plane at three redshifts; this is the same object viewed as a function of redshift. The model tracks the non-monotonic motion of the locus in the optical features and the steady decline of the mid-infrared excess. Model and data intervals nearly coincide here because at these magnitudes the measurement errors are small compared with the intrinsic spread; the deconvolution is a small correction in this projection and a large one at the faint end (Figure 2). Reproducing the observed spread after adding each object’s own errors back to draws from the model is a direct check that the fit is neither over- nor under-dispersed.

using quasars, costing roughly 1.3 in log BF — comparable to the entire same-redshift versus wrong-redshift separation. Spectroscopic quasars from DESI DR1 and SDSS DR16Q are therefore matched within $1''$ and removed.

This removes only the quasars somebody has *observed*; unobserved ones remain, a residual of the same bias in the same direction. They are meant to be carried by the field-quasar term through Σ_Q , which requires a spectroscopic completeness. That completeness matters less than it might appear, because Σ_Q multiplies λ_{same} and λ_{field} alike: a uniform factor-two error moves p_{same} by about 13%, identically for every candidate at a given magnitude, so the ranking is untouched. It is a *redshift-dependent* completeness that fails to cancel (§6).

The density depends on position, through Galactic structure and reddening, and on magnitude, through which populations are reachable at a given depth. We therefore fit a deconvolved mixture per (HEALPix cell, magnitude bin) and pool sparse cells towards coarser ones:

$$p = a_\ell p_{\text{cell}} + (1 - a_\ell) [a_p p_{\text{parent}} + (1 - a_p) p_{\text{global}}], \quad a = \frac{n}{n + n_0}, \quad (9)$$

with a pooling constant n_0 that is *intended* to be tuned on held-out sky blocks rather than fixed

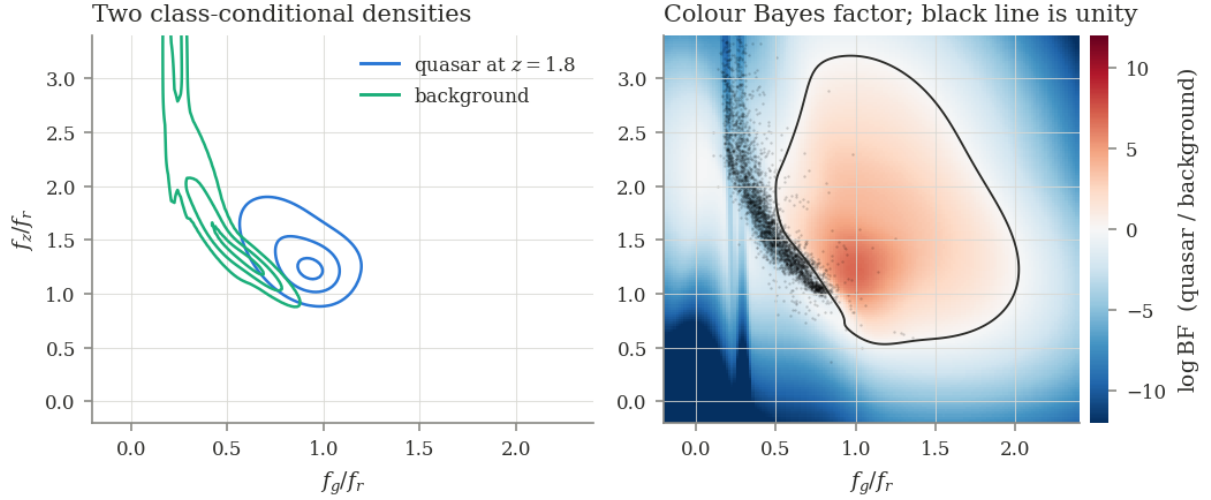


Figure 5: *Left*: the quasar density at $z = 1.8$ and the local background density, both fitted to real data, in the same feature plane. *Right*: the colour Bayes factor (5); the black contour is $\text{BF} = 1$, and black points are real catalogue sources at the relevant magnitude. Most of the field population sits where the Bayes factor strongly disfavours the quasar hypothesis, but the two loci are not disjoint.

in code. It is not yet: the tuning routine holds out whole cells at the same resolution as the parent level, so every validation object falls back to the global model and its score carries no information about n_0 . The figures in this note therefore use a fixed $n_0 = 500$, and the tuning must be repaired before any calibrated number depends on it. HEALPix is used in the nested scheme, so a parent cell is a bit shift, and in Galactic coordinates, which is the frame the structure lives in. Each scored object records the weight its own cell received, so a number that came from the pooled model is distinguishable from one that came from local data.

Two ways to be local, and a comparison between them. The hierarchy above amortises one set of fits over a whole footprint, which is what a survey-wide scan needs. For a handful of candidates the better answer is to fit the background in a cone around each one: the stellar population, the survey depth and the reddening are then the candidate’s own rather than a cell average, and the surveyed area is exactly πR^2 — which removes the commonest way of getting Σ_B wrong. At Legacy Surveys depth away from the plane the source density is of order $2 \times 10^4 \text{ deg}^{-2}$, so a radius of 0.5° supplies some 15,000 sources, ample for a mixture of this size while staying genuinely local.

Both are implemented, and the same candidates can be scored under both. They should agree where the field is smooth and the cell is well populated, and disagree where the local density, depth or reddening departs from the cell average. The comparison reports, per candidate, how far apart the two put the colour likelihood, the Bayes factor, R and the posteriors — so a result that depends on which background was assumed is visible as such rather than presented as a single number.

Figure 5 shows the two class-conditional densities in the same plane and the resulting colour Bayes factor. The populations overlap; the method does not pretend otherwise, and the Bayes factor is smoothly varying rather than a decision boundary.

6 Surface densities

The two surface densities must be measured over footprints whose selection and usable area are known, and the same cuts must be applied to the candidates. The illustration in this note does

not fully meet that standard: for speed it draws quasars from a 12° cone and background sources from a 1° cone at the same centre, and applies a mask-bit cut without a mask-corrected area. Each density uses its own area, so the arithmetic is consistent, but matched selection and area should be read as requirements of the method rather than as properties these particular figures demonstrate.

Background. $\Sigma_B(m, \ell, b)$ is estimated by counting the imaging catalogue into the same (cell, magnitude) bins and dividing by the *usable* area after masking. That area must be supplied; the implementation refuses to run without it. The reason is not pedantry: an earlier version of the figure pipeline defaulted to the nominal HEALPix pixel area, 53.7 deg^2 at nsid 8, for a background cone of 3.14 deg^2 , understating Σ_B by a factor of 17 and inflating every quasar posterior by the same factor. Cells that were surveyed but happen to hold no object in a magnitude bin must also stay in the denominator, or sparse bins are biased high. The masking is applied identically to the candidates, so the two are consistent by construction.

Quasars. $\Sigma_Q(z, m)$ requires more care, because a spectroscopic quasar catalogue’s redshift histogram is a targeting selection function, not a quasar population. We estimate

$$\Sigma_Q(z, m) = \frac{N(z, m)}{A \Delta z \Delta m C(z, m)}, \quad (10)$$

with A the surveyed area and $C \leq 1$ the spectroscopic completeness. The default is $C = 1$, which understates the quasar density everywhere.

That does *not* make p_{same} a lower bound, and an earlier version of this note wrongly said it did. The bound holds only if completeness is independent of redshift, so that correcting it scales λ_{same} and λ_{field} together and only the fixed background dilutes. If completeness is worse away from z_0 than at it, correcting it raises the field term more and the posterior *falls*: intensities $(1, 1, 1)$ give $1/3$, while a correction that multiplies only the field term by 100 gives $1/102$. $C = 1$ is a declared, reproducible default, not a conservative one. Where no defensible prior exists at all, the implementation returns the Bayes factor and leaves the posterior fields null with a status code, rather than substituting a class fraction.

7 The redshift window is a multiplicative constant

This section contains the main methodological point.

7.1 The tension

A physically motivated pair window is of order $\pm 2000 \text{ km s}^{-1}$. At $z_0 = 1.4$ that is

$$\Delta z = (1 + z_0) \frac{\Delta v}{c} = 2.4 \times \frac{2000}{299792.458} = 0.016. \quad (11)$$

Broadband colours constrain a quasar redshift far more loosely, and often to a multimodal distribution rather than a single peak — which is why XDQSOz produces a redshift *PDF* in the first place (Bovy et al., 2012). Measured on 1500 real DESI quasars with the model of §4, the median posterior width $\sigma_z \approx 0.6$ (16–84th percentile 0.36–0.78; 68% credible interval ≈ 0.71 wide). The median distance from the posterior mean to the spectroscopic redshift is 0.28, so the point estimate is better than the posterior width alone suggests; but it is the *width* that governs how much probability a narrow window can capture. Rather than quote a ratio — which would compare a standard deviation with a window width, two different summaries — note the two numbers directly: at $z_0 = 1.8$ the window spans 0.037 in redshift while the posterior has a standard deviation of 0.6. For a unimodal posterior of that width, a window of that span captures of order two per cent of the probability. Figure 6 shows this for three real quasars.

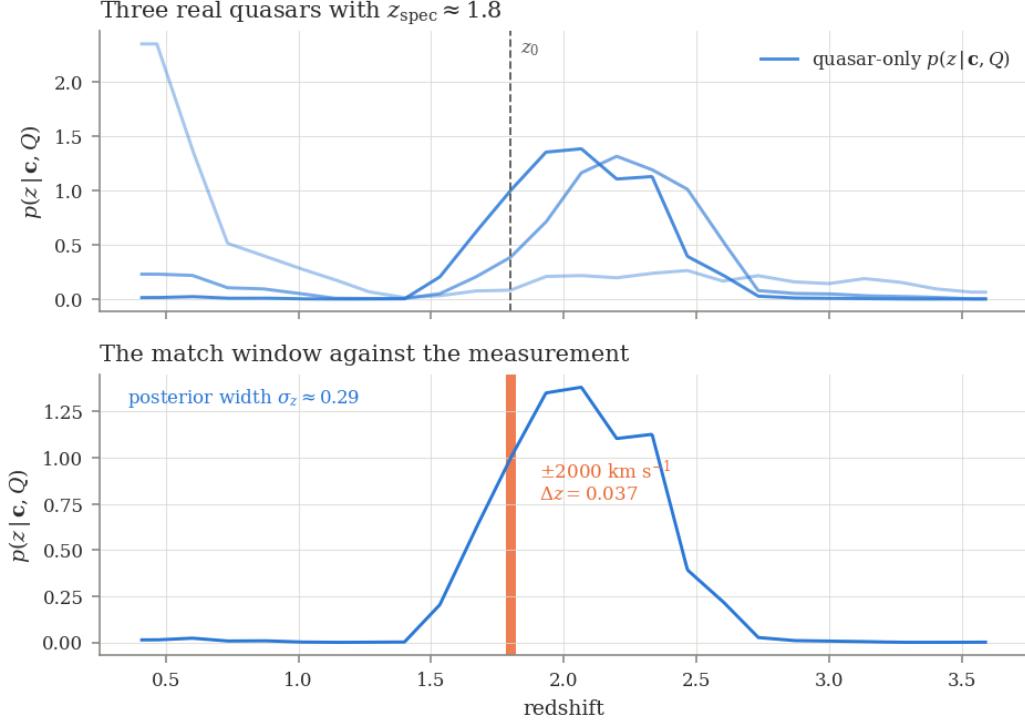


Figure 6: *Top*: quasar-only redshift posteriors $p(z | \mathbf{c}, Q)$ for three real DESI quasars with $z_{\text{spec}} \approx 1.8$, showing the width and the secondary solutions that broadband colours cannot resolve. *Bottom*: the same posterior with a $\pm 2000 \text{ km s}^{-1}$ window drawn to scale. The window is the narrow orange band. The width quoted on the panel is for *this* object, which is better constrained than most; across 1500 quasars the median width is $\sigma_z \approx 0.6$ (§7). Note that this is the quasar-only posterior under a *flat* redshift prior; p_{same} integrates the same window against $\Sigma_Q(z, m) p(\mathbf{c} | Q, z)$ instead. The panel shows the scale of the mismatch between window and measurement, which is what makes p_{same} small, not the quantity itself.

The arithmetic consequence is unavoidable: almost all of a candidate’s quasar intensity falls outside the window, so $\lambda_{\text{field}} \gg \lambda_{\text{same}}$ and p_{same} is small even for an object sitting exactly on the locus at exactly the right redshift. This is not a defect of the method. It is the correct answer to the question as posed, and it means the question as posed is not the one colours can answer.

7.2 Which uncertainty is the problem

It is worth being explicit about *whose* redshift is uncertain, because the two quantities differ by three orders of magnitude and only one of them matters.

	Δz at $z = 1.8$	relative
primary’s spectroscopic error	0.0004	$1\times$
$\pm 2000 \text{ km s}^{-1}$ window (half-width)	0.019	$49\times$
companion’s photometric width	0.6	$1579\times$

The primary is a spectroscopic object: the median DESI DR1 quasar redshift error is 0.0004, or 43 km s^{-1} . It is known some 1600 times better than the companion, and contributes essentially nothing to the blur. The entire difficulty sits on the companion, which has no spectrum. A perfectly known primary redshift does not help, because it is not the uncertain quantity.

What the primary’s precision *does* buy is that $p(\mathbf{c} | Q, z_0)$ is evaluated at a sharp point rather than marginalised over a second unknown redshift. That is a real simplification, and it is what

makes the calculation tractable at all. The primary’s error enters by *marginalising* the window over it, which smooths the window’s edges while conserving its area; a top hat becomes a difference of normal CDFs. An earlier implementation instead replaced the half-width by $\sqrt{s^2 + \sigma_{\text{primary}}^2}$, which increased the area, so a less certain primary produced a larger expected number of same-redshift quasars — uncertainty about where the window sits cannot create companions.

One caveat for practice: the 43 km s^{-1} above is a *pipeline statistical* error. Quasar systemic redshifts derived from broad lines, and C IV in particular, carry known systematic offsets that are a non-trivial fraction of a pair window. Where the window matters, `z_primary_err` should be set to a systemic uncertainty measured for the redshift estimator actually used, not to the pipeline value.

7.3 What sets the companion’s width

To first order,

$$\sigma_z \approx \frac{\text{scatter of colour at fixed } z}{|d(\text{colour})/dz|}, \quad (12)$$

a ratio between how fast the locus moves and how wide it is. Measured on our quasars near $z = 1.8$:

colour	$ dC/dz $	scatter at fixed z	implied σ_z
$g - r$	0.356	0.219	0.62
$r - z$	0.402	0.229	0.57

Combining the two as if independent gives 0.42; the full posteriors give 0.60, the difference being the correlation between colours and the multimodality created where the locus doubles back on itself.

The practical reading of (12) is that the denominator, not the numerator, is the lever. At bright magnitudes the scatter is dominated by *intrinsic* quasar-to-quasar diversity rather than measurement error — which is exactly what the injection test of Figure 2 establishes, with an intrinsic $\sigma_0 = 0.119$ in f_g/f_r where the measurement error is negligible. Deeper imaging in the same bands therefore buys little. Improving the redshift constraint requires bands in which quasar colour changes *faster* with redshift: a u band spanning the Lyman break, or a narrow band tracking a strong emission line.

7.4 Factorising the window out

The resolution is to stop treating the window as part of the model. Because it is much narrower than the scale on which $\Sigma_Q(z, m) p(\mathbf{c} | Q, z)$ varies, (1) becomes

$$\lambda_{\text{same}} \simeq \Delta Z_{\text{eff}} \Sigma_Q(z_0, m) p(\mathbf{c} | Q, z_0), \quad \Delta Z_{\text{eff}} \equiv \int W(z | z_0) dz, \quad (13)$$

with an error of order $(\Delta z / \sigma_z)^2$. The window area ΔZ_{eff} is $2s$ for a top hat of half-width s and $\sqrt{2\pi} s$ for a Gaussian, and is evaluated in closed form.

The step that makes this clean is the denominator. Since H_{same} and H_{field} *partition* the quasar intensity,

$$\lambda_{\text{same}} + \lambda_{\text{field}} + \lambda_{\text{bkg}} = \Lambda_{Q, \text{tot}} + \lambda_{\text{bkg}} \quad \text{for any window,} \quad (14)$$

where $\Lambda_{Q, \text{tot}} = \int \Sigma_Q(z, m) p(\mathbf{c} | Q, z) dz$. Defining

$$R \equiv \frac{\Sigma_Q(z_0, m) p(\mathbf{c} | Q, z_0)}{\Lambda_{Q, \text{tot}} + \lambda_{\text{bkg}}} \quad [\text{redshift}^{-1}] \quad (15)$$

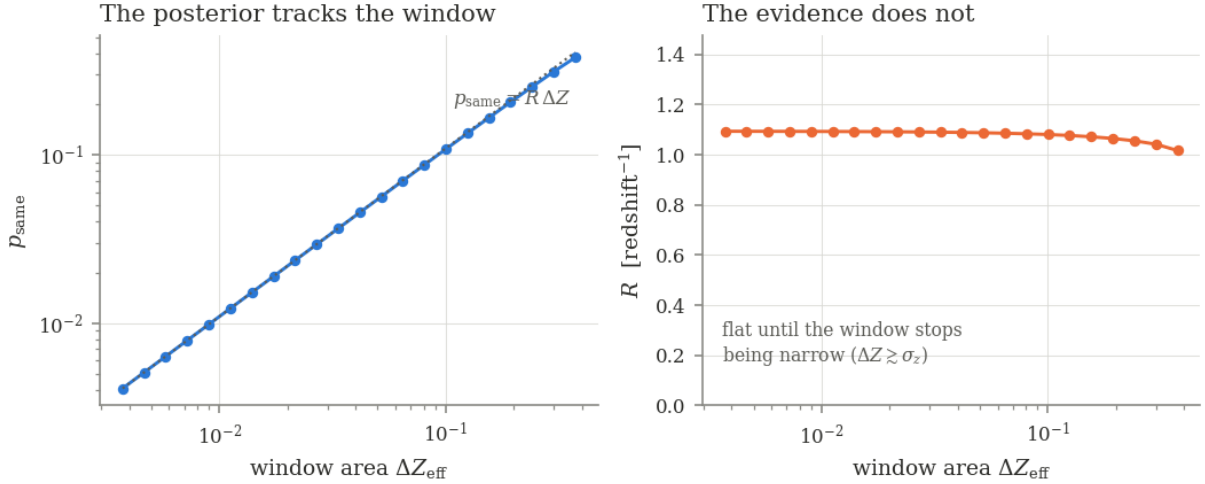


Figure 7: The factorisation, on a real quasar. *Left:* p_{same} is proportional to the window area over more than a decade, exactly as (16) requires; the dotted line is $R \Delta Z$ with R taken from the narrowest window. *Right:* R itself is flat, departing only when the window stops being narrow compared with σ_z and the approximation in (13) is no longer appropriate.

and combining (13)–(14) with (4) gives

$$p_{\text{same}} = R \Delta Z_{\text{eff}}, \quad (16)$$

a linear relation, not an odds transformation.

Two cautions about what this does and does not establish, both of which an earlier version of this note got wrong.

First, (14) is exact, but (13) is an approximation. Writing $f(z) = \Sigma_Q(z, m)p(\mathbf{c} | Q, z)$, the exact statement is

$$p_{\text{same}} = R(z_0) \Delta Z_{\text{eff}} + \frac{\int W(z) [f(z) - f(z_0)] dz}{D}, \quad D = \int f dz + \lambda_{\text{bkg}}, \quad (17)$$

so the linear relation holds to the extent that f is flat across the window. The leading relative correction is $f''(z_0)s^2/[6f(z_0)]$ for a top hat. Calling that $O((s/\sigma_z)^2)$ assumes the posterior width measures local curvature, which a multimodal posterior need not satisfy; at a cusp — and both the likelihood and the prior are interpolated piecewise-linearly — the error is instead $O(s)$.

Second, in the implementation R is computed as the *window-averaged* intensity, $R = p_{\text{same}}/\Delta Z_{\text{eff}}$. Equation (16) therefore holds for the reported number *by construction*; it is a definition, not a derived result. The substantive claim is the separate one that R does not depend on the window, and that holds only in the narrow limit. Figure 7 tests it on a real object: over a factor of ten in window width R moves by well under a per cent. For a window comparable to σ_z it is a genuine average and does vary.

7.5 Consequences

1. **Rank on R .** It is the window-free evidence, so two people can compare candidates without first agreeing on a window, and it is not compressed towards zero the way p_{same} is.
2. **Never quote p_{same} without its window.** The two travel together in every output row, as `p_sameq` and `dz_match_eff`.
3. **Never integrate a velocity window on the main redshift grid.** $\Delta Z_{\text{eff}} = 0.03$ on a grid of step 0.01 spans two or three points, and a sampled top hat there quantises the

answer in units of the grid step — up to 20% in the regime this project works in. An earlier implementation switched between that integral and the closed form (13) on a grid-resolution test, which merely chose between two inaccurate branches and also counted window area lying outside the trained redshift range. The window integral is now done on its own dense sub-grid, over the intersection of the window with the model’s support: one path, exact at any width, with no invented area at the edges.

4. **Widening the window does not improve the answer.** It changes the question. With R reported separately there is no incentive to.

7.6 How much discrimination is actually available

Figure 8 evaluates R against a primary at $z_0 = 1.8$ for three real populations, and the result is worth stating plainly because it sets expectations for the whole programme.

population	median $\log R$
quasars at $z \approx z_0$	−0.3
quasars at other redshifts	−2.6
random catalogue sources	−14.3

Two very different gaps. Against the field population the statistic is decisive: $e^{14.0} \approx 10^6$. Against *quasars at the wrong redshift* it is worth a factor of about 10.

That asymmetry is not a defect of the estimator; it is a statement about the information content of $grz + W1W2$ photometry. These bands identify quasars very well and locate them in redshift only weakly — which is visible directly in Figure 3, where the conditional loci at $z = 0.9$, 1.8 and 3.0 overlap substantially, and in the posterior widths of Figure 6.

The consequences for practice are concrete:

- **What this method delivers reliably is a quasar classification**, not a redshift match. A companion with high R is very likely a quasar; whether it is at z_0 is a much weaker inference.
- **The field-quasar hypothesis is therefore not a refinement but the dominant competing explanation**, which is precisely why a quasar-versus-star calculation would be misleading here rather than merely incomplete.
- **Improving redshift discrimination requires more information, not more training data.** This was originally stated as an expectation from (12). It has since been tested directly. Retraining on the full southern sample — 1,237,336 quasars against 79,865, with 32 redshift slices against 24 and twice the mixture components — leaves the redshift posterior unchanged:

training set	median σ_z	16–84%	median $ z_{\text{post}} - z_{\text{spec}} $
79,865, $K = 6$	0.601	0.36–0.75	0.290
1,237,336, $K = 12$	0.602	0.35–0.76	0.296

The comparison favours the larger model if anything: the small one was trained on a 452 deg^2 cone, so about 95% of the 1500 test quasars lay outside its training set, while all of them lay inside the large one’s. Fifteen times the data, evaluated with every advantage, moved σ_z by 0.001. The redshift precision is therefore set by the information in $grz + W1W2$, not by the size of the training sample or the flexibility of the model. What would help is a band in which quasar colour changes faster with redshift — a u band spanning the Lyman break,

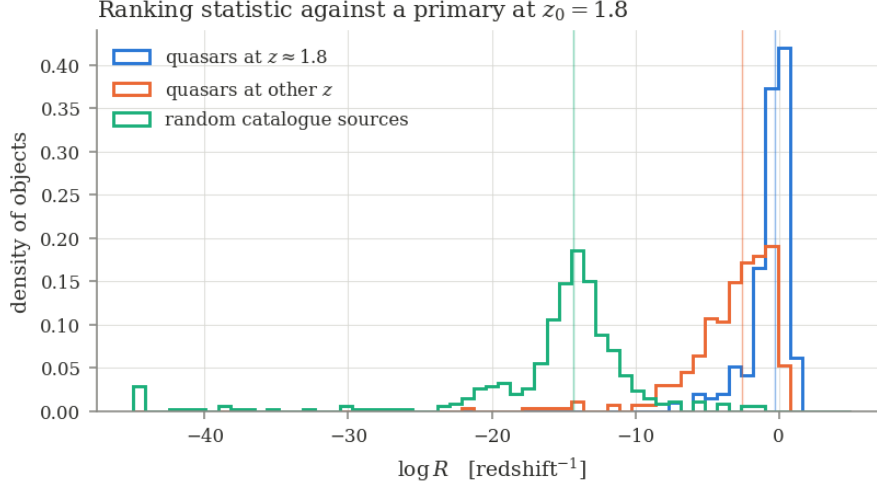


Figure 8: The ranking statistic evaluated against a primary at $z_0 = 1.8$, for real quasars at that redshift ($|z - z_0| < 0.05$), real quasars at other redshifts ($|z - z_0| > 0.6$), and random catalogue sources. Vertical lines mark the medians. Two limits of this comparison: the “same” class extends to 2.7 times the window half-width, so not all of it lies inside the window being scored; and nothing populates the gap between 0.05 and 0.6, which is where discrimination is genuinely hardest; and these objects were used in training. The quasar populations separate from the field by ~ 14 natural logarithms but from *each other* by only ~ 2.3 : *grz* + *W1W2* photometry identifies quasars far better than it locates them in redshift (§7.6). The figure shows the statistic’s dynamic range, not an operating characteristic.

a narrow band on a strong line — or variability, or spectroscopy. The modular likelihood structure is designed so such a term can be added as a factor rather than by rewriting the inference.

This bears on the information content, not on classification performance: the separations quoted above remain in-sample medians with the hardest cases absent, and establish neither purity nor completeness.

8 What the scorer returns

Every candidate produces a record separating evidence from posterior, with a status code and quality flags rather than a silent number.

<code>loglike_qso_zprimary</code>	$\log p(\mathbf{c} \mid Q, z_0)$; a likelihood, not a probability
<code>loglike_bkg</code>	$\log p(\mathbf{c} \mid B, m, \ell, b)$
<code>log_bayes_factor_qz_bkg</code>	(5); independent of both surface densities
<code>p_zmatch_given_qso</code>	$P(z \in W \mid \mathbf{c}, Q)$; conditional on being a quasar at all
<code>z_phot_mode</code>	mode of the quasar-only redshift posterior
<code>log_r_per_unit_z</code>	$\log R$ from (15); the ranking statistic, window-free in the narrow limit (§7)
<code>dz_match_eff</code>	ΔZ_{eff} <i>actually used</i> , i.e. the window area inside the model’s support, so that $p_{\text{same}} = R \Delta Z_{\text{eff}}$
<code>log_lambda_sameq, _fieldq, _bkg</code>	(1)–(3) in logs
<code>p_sameq</code>	(4), the three-hypothesis posterior
<code>p_sameq_vs_bkg</code>	the two-hypothesis number; ignores field quasars
<code>qso_ood_sigma</code>	distance to the nearest training component, in σ
<code>background_local_weight</code>	a_ℓ from (9); zero means a pooled model
<code>background_density_level</code>	0 local, 1 parent, 2 global
<code>n_bands_used</code>	usable feature dimensions for this object
<code>status, quality_flags</code>	why a value is missing, and what is suspect about one that is not
identification	<code>candidate_id</code> , <code>primary_id</code> , <code>ref_mag</code> , <code>photometric_system</code> , <code>model_manifest_id</code> — so a number can be traced back to the model and the data that produced it

9 A worked example on ten real objects

Figure 9 applies the classifier to five spectroscopic quasars and five point-like field sources, and draws the densities each one is judged against. It is deliberately the hard case: **DECaLS bands only**, with $W1$ and $W2$ marginalised out exactly through the observed-dimension mask rather than dropped. WISE carries most of the quasar/star discrimination, so this shows what the optical colours alone can do.

Each object is treated as a separate problem. The quasars are drawn from the model’s *reserved* spatial blocks, so none was used in fitting, and the field sources come one from each of five independent random fields — not five from one patch, which would share a single stellar environment and be a random sample of nothing. Every object then gets its *own* local stellar model, fitted in a 0.5° cone centred on it with known quasars removed and the masked area taken out of the density. The panels show three contours: the quasar model at that object’s target redshift, the stellar model fitted globally across the southern footprint, and the stellar model local to that object.

	z_0	r	log BF	p_{same}
quasars	1.42	19.2	+6.4	5.4×10^{-2}
	2.23	20.3	+7.5	3.3×10^{-2}
	1.84	21.2	+4.0	9.5×10^{-3}
	0.62	21.1	+4.8	3.1×10^{-3}
	2.85	19.7	−0.6	2.6×10^{-5}
field	1.84	21.3	−12.2	1.2×10^{-9}
	0.62	19.5	−4.2	1.1×10^{-6}
	2.23	20.5	−7.1	1.5×10^{-7}
	0.62	19.3	−6.1	1.9×10^{-7}
	0.62	20.8	−3.8	1.8×10^{-6}

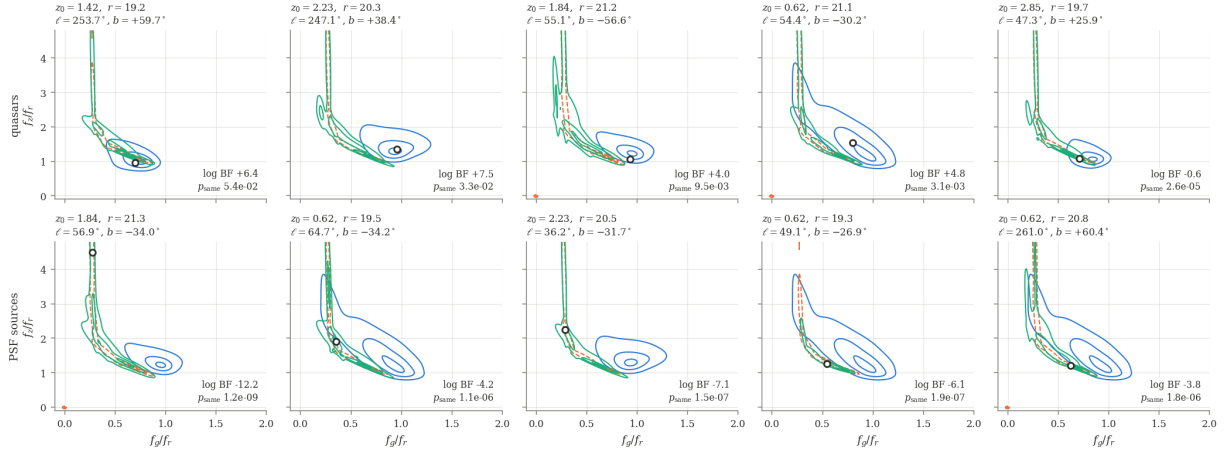


Figure 9: Ten real objects scored on DECaLS colours alone. *Top*: five quasars from the model’s reserved blocks. *Bottom*: five point sources, each from a different random field. Contours are the quasar model at z_0 (blue), the stellar model averaged over the footprint (orange, dashed) and the stellar model fitted in this object’s own neighbourhood (green). Panel titles give the target redshift, reference magnitude and Galactic coordinates; the measured object is the open circle with its error bars. Note that the outermost contour is drawn at 5% of peak density, *not* at an enclosed probability, so an object outside it is not thereby improbable.

Four of the five quasars separate cleanly, and all five field sources do. Three things in the figure are worth more than that headline.

The one failure is the expected one. The quasar at $z = 2.85$ returns $\log \text{BF} = -0.6$: its panel shows it sitting where the quasar locus crosses the stellar sequence. At that redshift the optical colours of a quasar and of a star coincide, and no amount of modelling recovers a distinction the photometry does not contain. This is precisely the degeneracy $W1 - W2$ exists to break, and the clearest single argument for not running optical-only in production.

The local and global stellar models agree here. Green and orange track each other closely in all ten panels. These sightlines are all at $|b| > 25^\circ$, where the field population is not varying rapidly, so the local fit *confirms* the footprint average rather than overturning it. That is a useful negative result, and one only visible because both were drawn. It should not be generalised to low latitude, where the per-object fit is expected to matter and has not been tested.

p_{same} is small even when the evidence is strong. The best object here has $\log \text{BF} = +7.5$ — odds of about 1800 to 1 on colour alone — and still only $p_{\text{same}} = 0.03$, because the $\pm 2000 \text{ km s}^{-1}$ window captures a few per cent of a redshift posterior that is an order of magnitude wider (§7). The Bayes factor and the ranking are what carry the information; the posterior carries the window.

A known defect visible in the figure. Small contour fragments appear near the origin in several panels, and a stray tick near (0.3, 4.6) in one. These belong to the *global* stellar model and mark a mixture component sitting where essentially no real sources are. They do not affect the local models or any number in the table, but they indicate a stray component in that fit which has not been chased down.

10 Validation design

Nothing above is evidence that the method works. That requires a labelled sample, and one exists: pairs in which *both* members have spectra. Around each spectroscopic quasar we take every companion that also has a spectrum and label it by what that spectrum says — a quasar at the same redshift, a quasar at a different redshift, or something else. The *labels* are spectroscopic and so independent of the colours being scored. The *inclusion* of an object in the sample is not: DESI targets quasars with a classifier built on these very photometric features, so the labelled set is a colour-selected subset of the sky. Stratifying by magnitude and latitude does not correct targeting, fibre assignment or redshift-success selection, and calibration measured on this sample transfers only to candidates drawn the same way. Quantifying that is part of the validation, not a caveat to be noted and passed over.

From DESI DR1, at separations of 3–20'', this yields approximately 1600 confirmed same-redshift pairs and 8400 projected quasar–quasar pairs, plus every confirmed star or galaxy companion. Calibration must then be assessed *stratified* by separation, blending diagnostic, magnitude, Galactic latitude and redshift, because the prevalence of each class changes strongly along all of them.

Two cautions were established while assembling this sample.

De-duplicate by position, not by identifier. Of the 10 636 DESI quasar “pairs” closer than 3'', 9577 lie within 0.5'' with identical redshifts: they are the same physical object entered twice under two target identifiers. The catalogue’s own primary-object flag does not remove them. Training on such a sample silently up-weights repeated objects.

Training-set selection is not neutral. DESI quasar targeting uses a random forest on the very Legacy Surveys colours we model, so the training set gives $p(\mathbf{c} \mid Q, z, \text{selected by DESI})$. The mitigation is to report held-out likelihood separately on an independently selected sample — SDSS DR16Q (Lyke et al., 2020), whose targeting channels differ — and to treat a large gap as a selection-bias warning rather than a curiosity.

11 Scope and limitations

Blending is the binding constraint. Below a few arcseconds the Legacy Surveys model fit divides flux between overlapping sources, so the two colour vectors are neither independent nor individually reliable, and a probability computed from them describes the deblender rather than the sky. The current pipeline therefore scores cleanly deblended companions only, enforced by an explicit separation floor and blending limit; a candidate lacking those measurements is treated as blended, because it cannot be certified clean without them. Closer pairs require image-level forced photometry, which is separate work.

p_{same} is not the probability of a physical pair. It is a classification posterior under the adopted population priors. A physical-pair calculation additionally requires conditioning the quasar intensity on angular separation from the primary through small-scale quasar clustering (Hennawi et al., 2006; Eftekharzadeh et al., 2017). That layer is deliberately separate, must come from an externally specified measurement used inside its documented regime, and must be reported separately from the photometric posterior.

Unused evidence. Gaia parallax and proper motion settle the question outright for bright companions and are available in the same catalogue; WISE photometry is the single strongest star/quasar discriminant but has a $\sim 6''$ beam and is therefore the most blend-contaminated.

Running optical-only and optical+WISE models side by side makes their disagreement a blending diagnostic. Variability is not uniformly available over the footprint and is out of scope here.

A Gaussian identities used

For a mixture (7) over a partitioned vector $\mathbf{v} = [\mathbf{a}, \mathbf{b}]$ and an observation $\mathbf{b} = \mathbf{y}$ with covariance $\mathbf{S}_y$, the conditional is again a mixture, with weights

$$w_k \propto \alpha_k \mathcal{N}(\mathbf{y} \mid \boldsymbol{\mu}_{b,k}, \mathbf{V}_{bb,k} + \mathbf{S}_y), \quad (18)$$

and components

$$\boldsymbol{\mu}_{a|y,k} = \boldsymbol{\mu}_{a,k} + \mathbf{V}_{ab,k}(\mathbf{V}_{bb,k} + \mathbf{S}_y)^{-1}(\mathbf{y} - \boldsymbol{\mu}_{b,k}), \quad (19)$$

$$\mathbf{V}_{a|y,k} = \mathbf{V}_{aa,k} - \mathbf{V}_{ab,k}(\mathbf{V}_{bb,k} + \mathbf{S}_y)^{-1}\mathbf{V}_{ba,k}. \quad (20)$$

The appearance of $\mathbf{S}_y$ in the weights is how an uncertain primary redshift broadens the match rather than being ignored. Marginalising over a subset of dimensions simply deletes the corresponding rows and columns. In the implementation no inverse is ever formed: every quadratic form and log determinant comes from a Cholesky factor, and all densities are computed in log space with `logsumexp`.

B The extreme-deconvolution update

For observation i with observed sub-vector $\mathbf{w}_i$, noise covariance $\mathbf{S}_i$, and component j , write O for the observed dimensions and

$$\mathbf{T}_{ij} = \mathbf{V}_j^{OO} + \mathbf{S}_i, \quad q_{ij} \propto \alpha_j \mathcal{N}(\mathbf{w}_i \mid \boldsymbol{\mu}_j^O, \mathbf{T}_{ij}), \quad (21)$$

$$\mathbf{b}_{ij} = \boldsymbol{\mu}_j + \mathbf{V}_j^{O\cdot} \mathbf{T}_{ij}^{-1}(\mathbf{w}_i - \boldsymbol{\mu}_j^O), \quad \mathbf{B}_{ij} = \mathbf{V}_j - \mathbf{V}_j^{O\cdot} \mathbf{T}_{ij}^{-1} \mathbf{V}_j^{O\cdot}, \quad (22)$$

with the M step

$$\alpha_j = \frac{1}{N} \sum_i q_{ij}, \quad \boldsymbol{\mu}_j = \frac{\sum_i q_{ij} \mathbf{b}_{ij}}{\sum_i q_{ij}}, \quad \mathbf{V}_j = \frac{\sum_i q_{ij} [\mathbf{B}_{ij} + (\mathbf{b}_{ij} - \boldsymbol{\mu}_j)(\mathbf{b}_{ij} - \boldsymbol{\mu}_j)^\top]}{\sum_i q_{ij}}. \quad (23)$$

With $\mathbf{S}_i = 0$ and no missing dimensions this reduces exactly to ordinary Gaussian-mixture EM. Restricting O per object is how missing bands are handled without projection matrices. Sample weights multiply every sum, which is where selection-function weights would enter.

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